

Uvedomíme, že  $[\text{Ker } \beta^i]_{\beta} = \text{Ker } [\beta^i]_{\beta}^{\beta} = \text{Ker } [\beta]_{\beta}^{\beta^i}$

$$a \quad [\text{Im } \beta^j]_{\beta} = \text{Im } [\beta^j]_{\beta}^{\beta} = \text{Im } [\beta]_{\beta}^{\beta^j} \quad \left[ [\beta]_{\beta}^{\beta^k} = 0 \text{ } k \geq 6 \right]$$

Matice  $[\beta]_{\beta}^{\beta} = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \end{pmatrix}$  a tedy mocniny budou  $[\beta]_{\beta}^{\beta^5} = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \end{pmatrix}$

→ toho jak vypadají  $[\beta]_{\beta}^{\beta^i}$  vidíme:

$$[\beta]_{\beta}^{\beta^2} = \begin{pmatrix} 0 & 0 & 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \end{pmatrix} \quad [\beta]_{\beta}^{\beta^3} = \begin{pmatrix} 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots \end{pmatrix}$$

$$\text{Ker } \beta = \text{LO}\{b_1, b_6, b_9\}$$

$$\text{Ker } \beta^2 = \text{LO}\{b_1, b_2, b_6, b_7, b_9, b_{10}\}$$

$$\text{Ker } \beta^3 = \text{LO}\{b_1, b_2, b_3, b_6, b_7, b_9, b_{10}\}$$

$$\text{Ker } \beta^4 = \text{LO}\{b_1, \dots, b_4, b_6, \dots, b_{10}\}$$

$$\text{Ker } \beta^5 = \text{LO}\{b_1, \dots, b_{10}\}$$

$$\text{Ker } \beta^6 = \dots$$

$$\vdots$$

$$\text{Im } \beta = \text{LO}\{b_2, b_3, b_4, b_5, b_7, b_8, b_{10}\}$$

$$\text{Im } \beta^2 = \text{LO}\{b_3, b_4, b_5, b_8\}$$

$$\text{Im } \beta^3 = \text{LO}\{b_4, b_5\}$$

$$\text{Im } \beta^4 = \text{LO}\{b_5\}$$

$$\text{Im } \beta^5 = \text{LO}\{\emptyset\}$$

$$\text{Im } \beta^6 = \dots$$

$$\vdots$$

$$[\beta^3]_{\beta} = \begin{pmatrix} 0 & 0 & 0 & 1 & 0 & \dots \\ 0 & 0 & 0 & 0 & 1 & \dots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \end{pmatrix}$$

dimenze:

| $j \setminus i$ | 1    | 2    | 3    | 4    | $\geq 5$ |
|-----------------|------|------|------|------|----------|
| 1               | 1, 0 | 1, 3 | 1, 5 | 6    | 7        |
| 2               | 1, 0 | 1, 0 | 1, 2 | 1, 3 | 1, 4     |
| 3               | 1, 0 | 1, 0 | 1, 0 | 1, 1 | 1, 2     |
| 4               | 1, 0 | 1, 0 | 1, 0 | 1, 0 | 1, 1     |
| $\geq 5$        | {0}  | {0}  | {0}  | {0}  | {0}, 0   |

$\text{Im } f^i \cap \text{Ker } f^j$   
Báze každého z prostorů bude počínáme  $(b_1, \dots, b_{i0})$  a bude složena

přes z tech vektorů  $b_1, \dots, b_{i0}$  které lze jeon na stejném  
místě jako  $\text{Im } f^i$  a zároveň  $\perp \text{Ker } f^j$

Pro  je to možné popsat